AST 6215 Stars and the Galaxy 2026 Problem Set #1

Due Sep 1, 2026 4 am

Useful numbers:
Solar M 1.989×10^{33} g Solar L 3.844×10^{33} erg/s Solar R 6.96×10^{10} cm

Please list all collaborators you worked with as well as all resources you referenced.

1. The nuclear burning lifetime of a star can be expressed as
$\tau_{nuc} = f \varepsilon M c^2 / L$
where f= fraction of fuel burned
ε= efficiency of matter to energy conversion
M= mass
L = luminosity
c = speed of light

Assuming $L \sim M^3$; $f \sim 0.1$; M~R; (If the units bother you, read '~' as "scales as")
Determine:
a) What is ε (the efficiency) for the conversion of 4 hydrogen atoms into one helium atom?
b) What is the nuclear burning lifetime of the sun?
c) What mass and temperature of star has a lifetime of ~ 1 Gyr?
d) What mass and temperature of star has a lifetime of ~ 0.1 Gyr?

2. Show that the equation of state for a photon gas is $P_{rad} = a/3\ T^4$. Note that you cannot start with a blackbody function, you should start from a particle approximation as we did in class.
You may find it helpful to work in frequency ν rather than momentum p.
You also might find it helpful to know that $\int_0^\infty \frac{u^3}{e^u - 1} du = \frac{\pi^4}{15}$

3. a) Write down the six equations needed to find the Saha balance of a hydrogen- helium mixture (you don't need to solve them, just write them all out)
b) Would you expect hydrogen to be more ionized or less ionized in a K giant versus a K dwarf of the same effective temperature? Justify your reasoning.

4. Calculate values for and compare the importance of gas pressure and radiation pressure at
a) the core of the Sun
b) the surface of the Sun
You are welcome to make simplifying assumptions like we did in class, but please make them clear. Interpret what this means for our future calculations of stellar interiors and stellar atmospheres.

5. Use the Yale Rotating Evolution Code (YREC, https://github.com/yreclab/yrec, Pinsonneault et al. 2026) to run a 1 Solar Mass, solar composition model to the end of core hydrogen burning (Terminal Age Main Sequence).
a) Show a plot of luminosity versus temperature (HR diagram) for this model
b) What is the age at the end of core hydrogen burning? Does it differ from your answer in 1b, and if so, why? What assumptions do you think mattered the most?